THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 31	Add the two distances from the centre.
2	**(b)** 96	Use Brahmagupta's formula.
3	**(b)** $\frac{1}{9}$	List all pairs with the given product.
4	**(c)** 50	$d = 3$, $a = 5$; $t_{16} = a + 15d$.
5	**(a)** 17	Subtract the two distances from the centre.
6	**(c)** 42	Side of square $= 2r$.
7	**(a)** $\frac{2}{9}$	Count the favourable pairs out of 36.
8	**(c)** 37	$t_k = S_k - S_{k-1} = p(2k-1) + q$.
9	**(d)** 14	Add the distances of the centres from the common chord.
10	**(c)** 770	Difference of the two areas.
11	**(a)** $\frac{7}{8}$	Use the complement rule where it is quicker.
12	**(c)** 816	AP with $a = 6$, $d = 6$.
13	**(a)** Both A and R are true and R is the correct explanation of A.	Half-chord $= \sqrt{5^2 - 3^2} = 4$ cm, using the perpendicular bisection in R; chord $= 8$ cm.
14	**(a)** Both A and R are true and R is the correct explanation of A.	$s = 21$, area $= \sqrt{21 \times 8 \times 7 \times 6} = \sqrt{7056} = 84$, using R.
15	**(a)** Both A and R are true and R is the correct explanation of A.	$P = \frac{3}{6} = \frac{1}{2}$, obtained exactly as in R.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Distances: $\sqrt{25^2 - 7^2} = 24$ cm and $\sqrt{25^2 - 24^2} = 7$ cm (1); Distance $= 24 + 7 = 31$ cm (1).	1+1
17	$s = 23$ cm (1); $A = \sqrt{(s-a)(s-b)(s-c)(s-d)} = 96$ cm^2 (1).	1+1
18	Favourable pairs: (2,6), (3,4), (4,3), (6,2) (1); $P = \frac{4}{36} = \frac{1}{9}$ (1).	1+1
19	$\frac{n}{2}[2a + (n-1)d] = 255$; $d = 2$ (1); $t_{15} = 31$ (1).	1+1
20	Distances of centres from the chord: 6 cm and 15 cm (1); Distance $= 21$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$d_1 = \sqrt{13^2 - 12^2} = 5$ cm, $d_2 = \sqrt{15^2 - 12^2} = 9$ cm (1); Opposite sides: $5 + 9 = 14$ cm (1); Same side: $\lvert 5 - 9 \rvert = 4$ cm (1).	1+1+1
22	Radius $= \frac{44}{2\pi} = 7$ cm; Side $= \frac{44}{4} = 11$ cm (1); Circle area $= 154$ cm^2; Square area $= 121$ cm^2 (1); The circle encloses the larger area (1).	1+1+1
23	$P(\text{green}) = 1 - \frac{1+1}{3} = \frac{1}{3}$; Red $= 8$ (1); Blue $= 8$ (1); Green $= 8$ (1).	1+1+1
24	$2a + 10d = 40$ and $a + 4d = 17$ (1); Solving: $a = 5$, $d = 3$ (1); $S_{20} = 670$ (1).	1+1+1
	OR: Distances from the centre 12 cm and 5 cm (1); Same side: 7 cm (1); Opposite sides: 17 cm (1).	1+1+1
25	Boundary $= 2 \times 60 + 2\pi r$; $= 120 + 88 = 208$ m (1); Area $= 60 \times 28 + \pi r^2$ (1); $= 1680 + 616 = 2296$ m^2 (1).	1+1+1
	OR: $P(\text{prime}) = \frac{52}{100} = \frac{13}{25}$; $P(>4) = \frac{31}{100} = \frac{31}{100}$ (1); Either $= 52 + 31 - 14 = 69$ (1); $P = \frac{69}{100} = \frac{69}{100}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) The perpendicular from the centre bisects the chord, so the half-length is 12 m (1). (ii) $\sqrt{13^2 - 12^2} = 5$ m (1). (iii) Half-chord $= \sqrt{13^2 - 12^2} = 5$, so the chord is 10 m (1). (iv) Distances from the centre are 5 m and 12 m on opposite sides, so the distance between them is $5 + 12 = 17$ m (2).	1+1+1+2

27	Sum 6: 5 outcomes; more than 6: 21 (1); $P(\text{sum}=6)=\frac{5}{36}$ (1); $P(>6)=\frac{7}{12}$ (1); $P(\leq 6)=1-\frac{7}{12}=\frac{5}{12}$ (1); The two probabilities add to 1 (1).	1+1+1+1+1
	OR: Penalties form an AP (1); $t_{30}=200+29\times 50=1650$ (1); $S_n=\frac{n}{2}(a+l)$ (1); $S_{30}=\frac{30}{2}(200+1650)$ (1); Total ₹27750 (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 7	$\frac{n}{2}[2a+(n-1)d]=510$ gives d.
2	**(d)** 108°	Linear pair at D.
3	**(a)** 120	Diagonals bisect at right angles: the other diagonal is $2\sqrt{s^2-(d/2)^2}$.
4	**(c)** 21	Blue = total − red.
5	**(b)** 14	Take $m-d,\ m,\ m+d$; the sum gives the middle number.
6	**(d)** 132°	Opposite angles of a cyclic quadrilateral.
7	**(a)** 8	The shortest altitude stands on the longest side.
8	**(d)** $\frac{7}{11}$	Count with repetition.
9	**(b)** 12100	Yearly values form a GP with ratio $1+\frac{r}{100}$.
10	**(d)** 168	Area $=\frac{1}{2}\times$ chord $\times$ distance.
11	**(a)** 484	Side $=\frac{\text{perimeter}}{4}$.
12	**(d)** 600	Expected failures $=N(1-p)$.
13	**(a)** Both A and R are true and R is the correct explanation of A.	$5-2=8-5=11-8=3$; a constant common difference is the definition of an AP.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true theorems, but R does not by itself explain why equal chords are equidistant.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	$2\pi r=2\times\frac{22}{7}\times 7=44$, so A is true; R is true but is the area formula, not the reason for A.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Side of square = 14 cm, area = 196 cm^2 (1); Circle area = 154 cm^2; difference = 42 cm^2 (1).	1+1
17	Let blue $=x$: $\frac{x}{12+x}=\frac{1}{4}$; $4x=12+x$ (1); $x=4$ (1).	1+1
18	$a+(n-1)d>200$; $(n-1)>32.50$ (1); $n=34$ (1).	1+1
19	$\angle ABC=32^\circ+41^\circ=73^\circ$ (1); $\angle ADC=180^\circ-73^\circ=107^\circ$ (1).	1+1
20	$d_2=\frac{2\times120}{10}=24$ cm (1); Side $=\sqrt{5^2+12^2}=13$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Terms: $12, 17, 22, \ldots, 197$ (1); Number of terms = 38 (1); Sum $=\frac{n}{2}(a+l)=3971$ (1).	1+1+1
22	$\angle BCD=180^\circ-34^\circ=146^\circ$ (1); $\angle ADB=90^\circ$ (angle in a semicircle) (1); $\angle ABD=90^\circ-34^\circ=56^\circ$ (1).	1+1+1
23	$s=21$, $A=84$ cm^2 (1); Altitude on 13 cm $=\frac{168}{13}=\frac{168}{13}$ cm (1); Altitude on 14 cm $=\frac{168}{14}=12$ cm (1).	1+1+1
24	Number of letters = 11; $P(\text{not } S)=1-\frac{4}{11}=\frac{7}{11}$ (1); Vowel or S: 8 letters (1); $P=\frac{8}{11}$ (1).	1+1+1
	OR: Let them be $\frac{m}{r},\ m,\ mr$; Product $=m^3=216$, so $m=6$ (1); $\frac{6}{r}+6r=15$ gives $r=2$ (1); Numbers 3, 6, 12 (1).	1+1+1
25	$d_1=\sqrt{13^2-12^2}=5$ cm, $d_2=\sqrt{20^2-12^2}=16$ cm (1); Opposite sides: $5+16=21$ cm (1); Same side: $\lvert5-16\rvert=11$ cm (1).	1+1+1
	OR: Radius $=\frac{88}{2\pi}=14$ cm; Side $=\frac{88}{4}=22$ cm (1); Circle area = 616 cm^2; Square area = 484 cm^2 (1); The circle encloses the larger area (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $2\pi r=2\times\frac{22}{7}\times14=88$ m (1). (ii) $\pi r^2=\frac{22}{7}\times196=616$ m^2 (1). (iii) Outer radius = 21 m, area $=\frac{22}{7}\times441=1386$ m^2 (1). (iv) Path area $=1386-616=770$ m^2 (1); cost $=770\times40=$ Rs 30800 (1).	1+1+1+2

27	Let x be removed: $\frac{12-x}{27-x} = \frac{1}{4}$ (1); $4(12-x) = 1(27-x)$ (1); $48 - 4x = 27 - x$ (1); $x = 7$ (2).	1+1+1+2
	OR: Ratio $\frac{2}{3}$ is constant, so a GP (1); 1st rebound $= 54$ m (1); 3rd rebound $= h \times (2/3)^3$ (1); $= 24$ m (2).	1+1+1+2

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 0.80	Reason about probabilities.
2	**(d)** 440	Add the first and last term, multiply by $\frac{n}{2}$.
3	**(d)** 35	Add the two distances from the centre.
4	**(d)** 126	Use Brahmagupta's formula.
5	**(c)** $\frac{1}{9}$	List all pairs with the given product.
6	**(c)** -4	$d=-4$, $a=60$; $t_{17}=a+16d$.
7	**(b)** 5	Subtract the two distances from the centre.
8	**(d)** 168	Side of square $=2r$.
9	**(a)** $\frac{5}{12}$	Count the favourable pairs out of 36.
10	**(d)** 70	$t_k=S_k-S_{k-1}=p(2k-1)+q$.
11	**(c)** 21	Add the distances of the centres from the common chord.
12	**(b)** 2464	Difference of the two areas.
13	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; R is a general range, so it does not explain why an impossible event gets exactly 0.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	The ratio of consecutive terms of the first is 2 and the difference of consecutive terms of the second is 2; both are true, but they are unrelated statements.
15	**(c)** A is true but R is false.	A is the cyclic-quadrilateral theorem; R is false in general (e.g. angles 80° and 100°).

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$P(\text{no head})=P(\text{TTT})=\frac{1}{8}$ (1); $P(\text{at least one head})=1-\frac{1}{8}=\frac{7}{8}$ (1).	1+1
17	$t_1=S_1=7$ (1); $t_{10}=S_{10}-S_9=61$ (1).	1+1
18	Distances: $\sqrt{25^2-15^2}=20$ cm and $\sqrt{25^2-20^2}=15$ cm (1); Distance $=20+15=35$ cm (1).	1+1
19	$s=26$ cm (1); $A=\sqrt{(s-a)(s-b)(s-c)(s-d)}=126$ cm^2 (1).	1+1
20	Favourable pairs: (1,6), (2,3), (3,2), (6,1) (1); $P=\frac{4}{36}=\frac{1}{9}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$P(\text{green})=1-\frac{1+2}{5}=\frac{2}{5}$; Red $=8$ (1); Blue $=16$ (1); Green $=16$ (1).	1+1+1
22	$2a+10d=64$ and $a+4d=28$ (1); Solving: $a=12$, $d=4$ (1); $S_{14}=532$ (1).	1+1+1
23	Distances from the centre 15 cm and 8 cm (1); Same side: 7 cm (1); Opposite sides: 23 cm (1).	1+1+1
24	Boundary $=2\times80+2\pi r$; $=160+44=204$ m (1); Area $=80\times14+\pi r^2$ (1); $=1120+154=1274$ m^2 (1).	1+1+1
	OR: $P(\text{prime})=\frac{108}{200}=\frac{27}{50}$; $P(>4)=\frac{63}{200}=\frac{63}{200}$ (1); Either $=108+63-30=141$ (1); $P=\frac{141}{200}=\frac{141}{200}$ (1).	1+1+1
25	Terms: $24, 31, 38, \ldots, 297$ (1); Number of terms $=40$ (1); Sum $=\frac{n}{2}(a+l)=6420$ (1).	1+1+1
	OR: $\angle BCD=180^\circ-41^\circ=139^\circ$ (1); $\angle ADB=90^\circ$ (angle in a semicircle) (1); $\angle ABD=90^\circ-41^\circ=49^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) 6 outcomes: 1, 2, 3, 4, 5, 6 (1). (ii) $\frac{3}{6}=\frac{1}{2}$ (1). (iii) Favourable outcomes 5, 6, so $\frac{2}{6}=\frac{1}{3}$ (1). (iv) Multiples of 3 are 3 and 6, so $P=\frac{1}{3}$ (1); $P(\text{not})=1-\frac{1}{3}=\frac{2}{3}$ (1).	1+1+1+2

27	Sum 7: 6 outcomes; more than 7: 15 (1); $P(\text{sum}=7)=\frac{1}{6}$ (1); $P(>7)=\frac{5}{12}$ (1); $P(\leq 7)=1-\frac{5}{12}=\frac{7}{12}$ (1); The two probabilities add to 1 (1).	1+1+1+1+1
	OR: Penalties form an AP (1); $t_{24}=100+23\times 25=675$ (1); $S_n=\frac{n}{2}(a+l)$ (1); $S_{24}=\frac{24}{2}(100+675)$ (1); Total ₹9300 (1).	1+1+1+1+1

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